MASON TODD PRATHER

SOFTWARE & SYSTEMS ENGINEERING | APPLIED AI | RESEARCH & TECHNICAL DELIVERY

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PROFESSIONAL PROFILE

Software and systems engineer with industry and university research experience delivering data pipelines, APIs, interactive applications, and engineering visualization. Background spans programmable networking, robotics data collection, collaborative XR, and research operations. Combines hands-on development and agentic AI workflows with technical analysis, documentation, and testing.

TECHNICAL AND DELIVERY CAPABILITIES

Programming and integration Python, C#, C++, C, JavaScript, HTML/CSS, SQL, P4, Bash, Rust, R; .NET, gRPC, Protocol Buffers, REST APIs, WebSockets, socket messaging.

Data and reporting Telemetry and data pipelines, InfluxDB, Grafana, PCAP, dashboards, structured export; Excel, Power BI, Tableau.

Artificial intelligence Agentic workflows, AI-assisted development, prompt engineering, MCP tooling, machine learning model training and inference integration into applications.

Interactive and spatial systems Unity, Unreal Engine, OpenXR, Meta XR, multi-user VR, avatars, eye/facial tracking, spatial UI; ROS 2, NVIDIA Omniverse, Isaac Sim, CloudXR.

Engineering delivery Git, Linux, Docker, Cloudflare R2, workflow automation, requirements translation, stakeholder collaboration, QA, technical documentation, onboarding, and system handoff.

PROFESSIONAL EXPERIENCE

Prather Software | Founder / Software Engineer | Nov 2026 - Present

- Founded an engineering and publishing company; lead product scoping, implementation, integration, and testing for independent software, XR applications, and games in development.
- Built and deployed the company's Astro/TypeScript website on Cloudflare, using reusable content models and accessible navigation to present its services and product portfolio.

OPEX Systems | Software / Systems Engineer | Feb 2024 - Jul 2026

- Served as a major developer of CIREN-5G, a DoD-funded Intel Tofino software platform; the team completed the contract in June 2026.
- Developed P4 forwarding and packet-management logic, Python telemetry pipelines, C-based classification, APIs, database integration, frontend components, and engineering documentation.
- Built a C# / Python interface and data-collection package using gRPC and Protocol Buffers for an autonomous wireless-scanning robot. Mapped Wi-Fi, BLE, LTE, and 5G measurements to signal-strength heat maps for an MxD collaboration.
- Developed LiDAR- and camera-driven digital-twin functionality, integrating inference-based detection of people, objects, and scene geometry, plus camera-driven avatar posing.
- Created 3D visualizations of 5G testbeds for engineer onboarding and technical demonstrations. Integrated Omniverse, Isaac Sim, CloudXR, and sensor data into simulation and visualization workflows.

PROFESSIONAL EXPERIENCE CONTINUED

Kennesaw State University | Graduate Research Assistant / XR Developer | Aug 2024 - May 2026

- Built peer-to-peer Unity VR collaboration for research led by Dr. Lei Zhang, synchronizing full-body avatars and facial expressions across participants. Integrated browser-based local-network photo and video sharing from smartphones.
- Engineered automated Quest 2 recording, video-file preparation, and cloud-storage workflows for a Penn State research collaboration, reducing reliance on manual capture steps.
- Implemented interactive XR features for collaborative storytelling and medical visualization associated with IEEE VR 2025 and 2026 3DUI projects. Supported study builds, testing, documentation, and collaborator handoffs.

Kennesaw State University | Undergraduate Research Assistant / XR Developer | Apr 2022 - Aug 2024

- Delivered a Unity eye-tracking attention-training application used in research led by Dr. Joy Li, translating specialized study requirements into usable VR interactions.
- Completed customer-delivered VR cybersecurity training software and a Unity application used in depression research, including gameplay, UI, input handling, QA, and deployment.

SELECTED INDEPENDENT AND COLLABORATIVE PROJECTS

Attention-Aware VR Classroom. Master's thesis system using Unreal Engine and C++ for gaze-based attention, quiz interventions, and structured research logging.

Quest Theater. Independent Unity/Android theater prototype integrating PhairPlay screen mirroring, native decoding, and movable virtual displays.

Promenade. Mobile party game in development with six minigames. Independent work includes player and bot movement, scoring, touch UI, session flows, and iPhone playtesting.

Rick Dangerman Remastered. Single-player run-and-gun remaster of the original university project, inspired by Call of Duty Zombies. Independent development covers movement, combat, progression, checkpoints, enemy behavior, levels, and art integration. Original team: Mason Prather, Cameron Stevenson, Hayden Rhodus, and Patrick Reyman.

EDUCATION

Master of Science in Software Engineering | May 2026
Kennesaw State University

Bachelor of Science in Computer Game Design and Development | May 2024
Kennesaw State University

SELECTED PUBLICATIONS

Prather, M., & Jung, S. (2026). Use of Exoskeleton Chair for Supporting Posture Change in VR Experience. IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW). DOI: 10.1109/VRW70859.2026.00260

Afroz, S., Haynes, D., Freeman, R., Prather, M., et al. (2026). Beyond Slices: a Narrative-Driven, Multi-User, Multi-Modal Virtual Reality System for Medical Imaging Presentation. IEEE VRW. DOI: 10.1109/VRW70859.2026.00344

Zhang, L., Prather, M., Haynes, D., Lu, F., et al. (2025). Saving the Polar Bears: A Collaborative VR Storytelling Experience Enhanced by Multi-scale and Multi-view WiM Designs. IEEE VRW. DOI: 10.1109/VRW66409.2025.00418